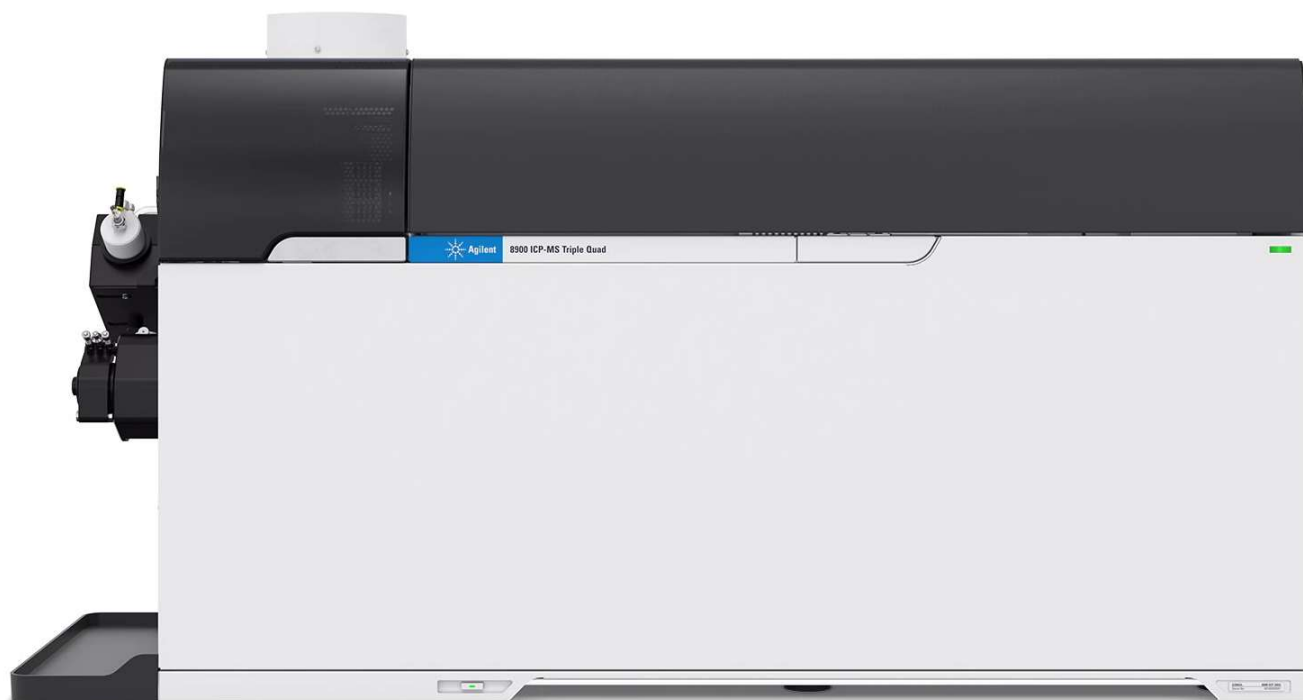


Agilent 8900 ICP-MS Triple Quadrupole User Guide



QUANTITATIVE BIO ELEMENT ANALYSIS AND MAPPING CENTER

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Pre-start checklist

Turn on the chiller

The chiller is located just outside the lab. Flip the switch on the chiller to turn it on.



Turn on the argon flow

The argon tanks are next to the chillers.



Check the tank gauge to be sure there is enough argon in the tank. If the tank is nearly empty, contact Keith MacRenaris (macrenar@msu.edu).

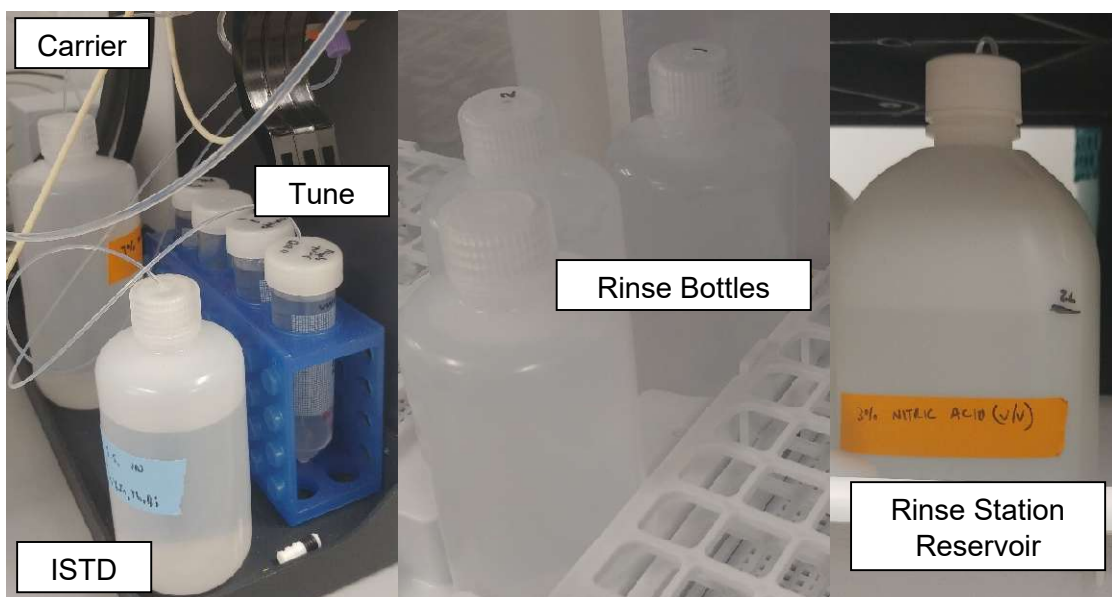


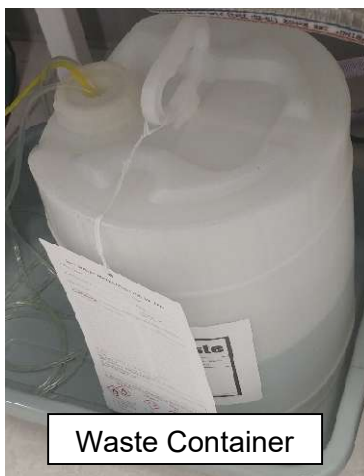
Make sure the correct regulator is attached to the argon tank so gas can flow to the instrument. The tubing should have tape labeled "MS" on it. If the correct regulator is attached, check the valve on the tank nozzle that is attached to the regulator by twisting counterclockwise. If it is fully open, it should give resistance when twisted. When fully opened, rotate back a half twist.



Check the liquids

Check to make sure the appropriate bottles are all sufficiently filled for the experiment. This includes: carrier, internal standard, and tuning solutions, and rinse station bottles. Also check to make sure the waste container below the instrument is not going to be overfilled during the experiment.





Waste Container

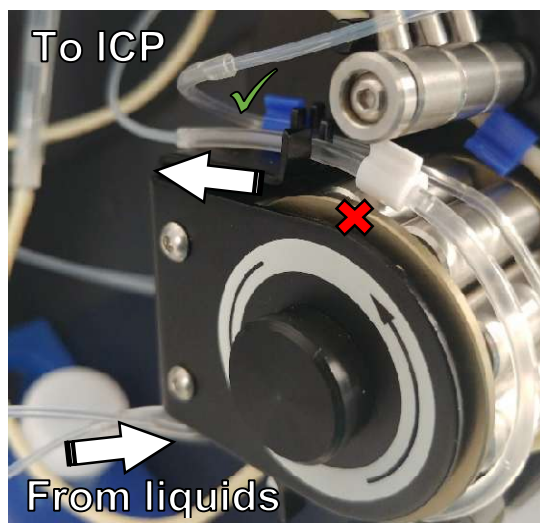
The carrier, internal standard, and tune solutions should all be located just to the left of the instrument. The rinse bottles are located in the middle of the autosampler and the rinse station reservoir is also to the left of the instrument, behind the other bottles (there is a tube running from the cap). The internal and tune solutions drain very slowly and generally will be filled by the staff. The carrier and rinse solutions are 3% v/v HNO_3 and can be filled using bottles available in the lab.

Clamp the tubing

Both the MS and Autosampler tubing must be pressed against their respective peristaltic pumps.

MS Tubing:

Make sure that the tubing is running parallel to the clamp. Stretch the tubing so that the tube is threaded through the notch and the clip is resting against the notch.



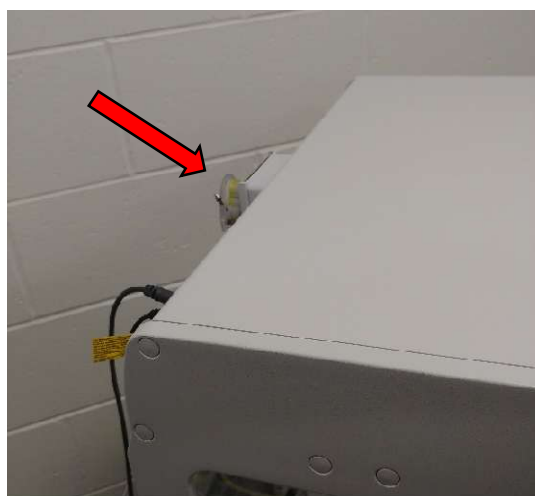
Note that the blue clip is resting against the notch (correct), but the white clip is not (incorrect). Also ensure the liquids are flowing towards the instrument.

Rotate the clamps so they are pressing against the tubes. Then rotate the tensioners so they are in the notch on the clamp. This will tighten the clamps against the tubes.

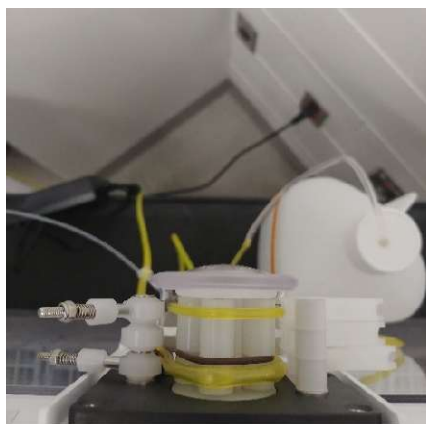


Autosampler tubing:

The peristaltic pump is located on the back of the autosampler.

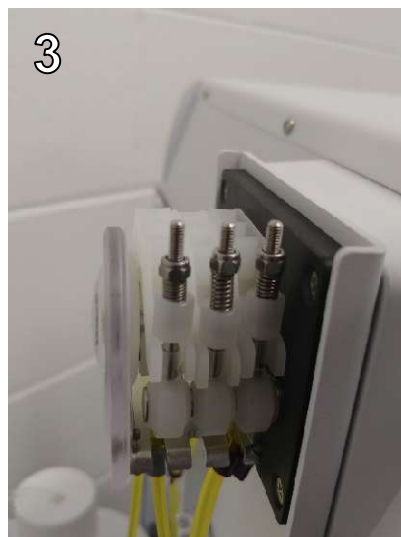
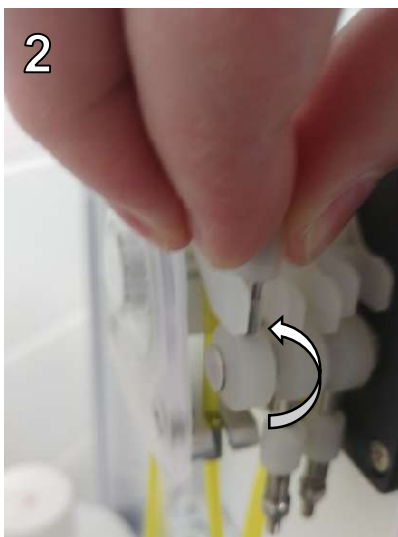


Make sure the tubes are running parallel to the clamps and stretch the tubes so that the clips rest against the bottom of the notch.



Note the gray clip is resting against the notch (correct), while the purple clip is loose (incorrect).

Place the clamps against the tubes. Rotate the tensioners up and into their slot on the clamp. Be sure to lift the bumper on the tensioner and release it once the tensioner is in the slot on the bracket.



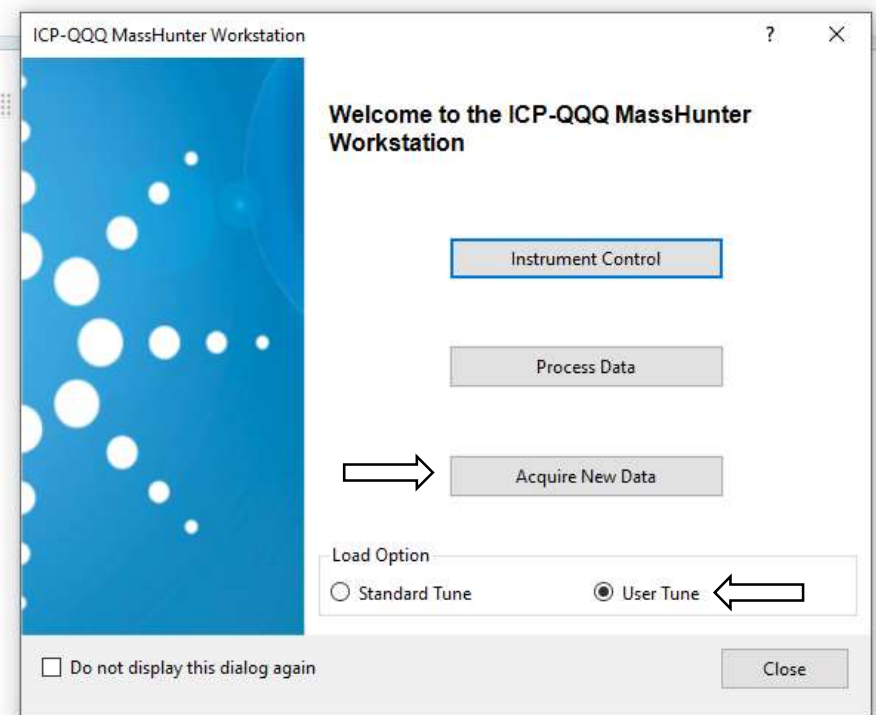
Starting the instrument

Opening the software

Select the ICP-MS Instrument Control software on the desktop.

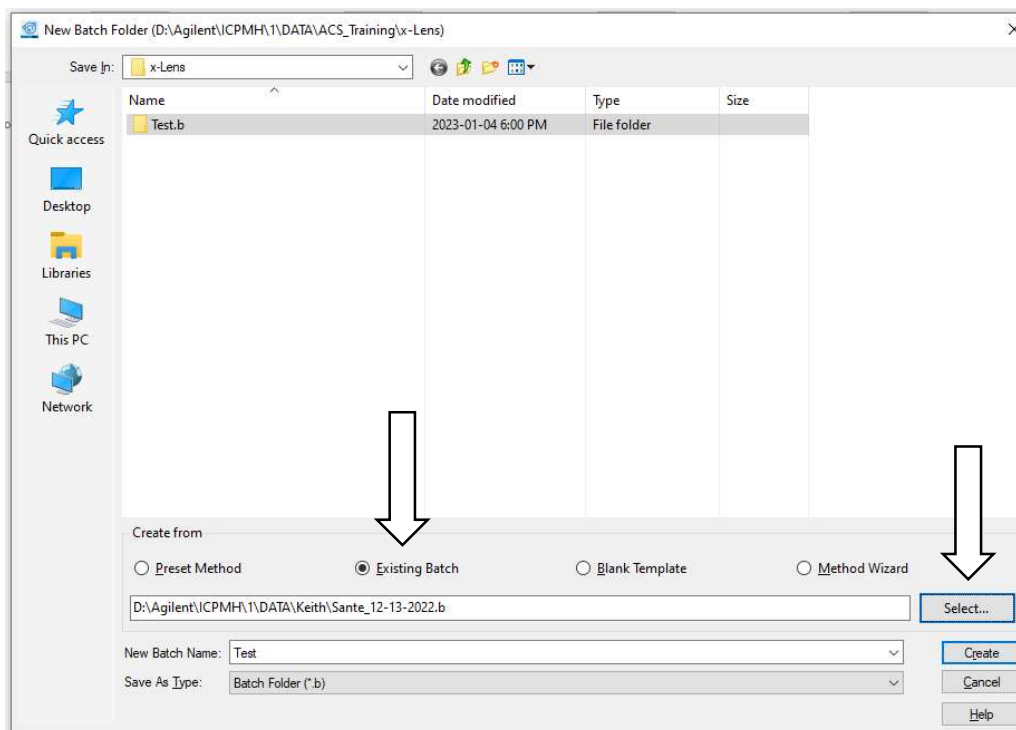


The program has three options – for general use, select “Acquire New Data,” with User Tune.

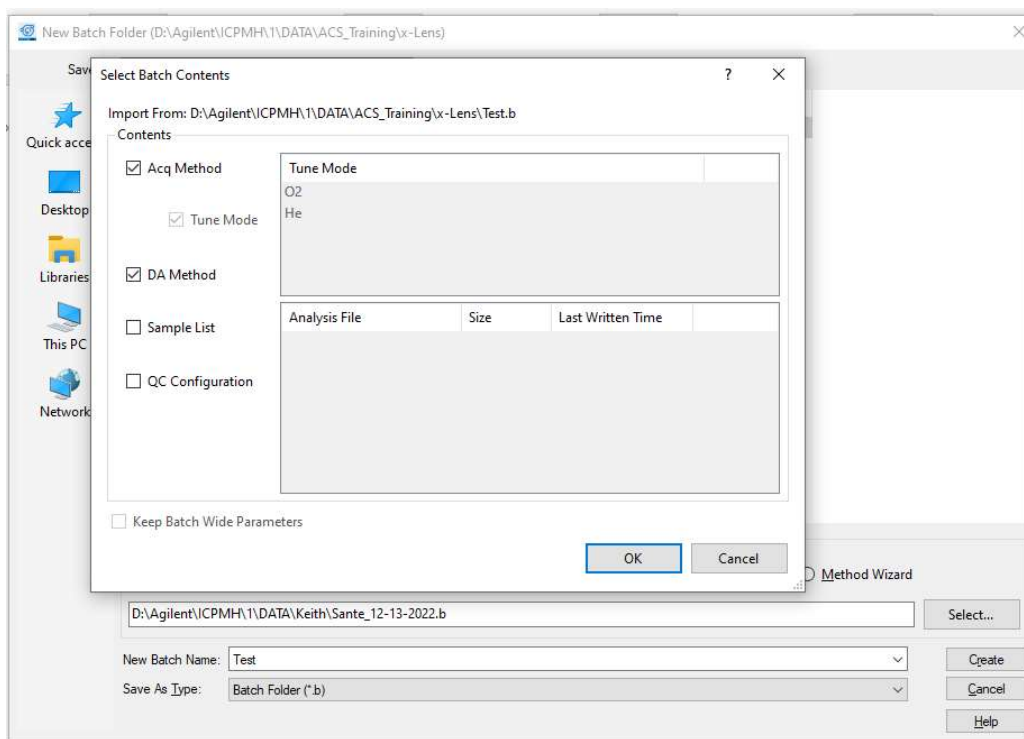


Creating a new batch

From the new batch screen, it is highly recommended to create a new batch using settings from an existing batch. If this is the first time creating a batch, ask Keith or Aaron to select an appropriate batch template for the needs of your experiment.

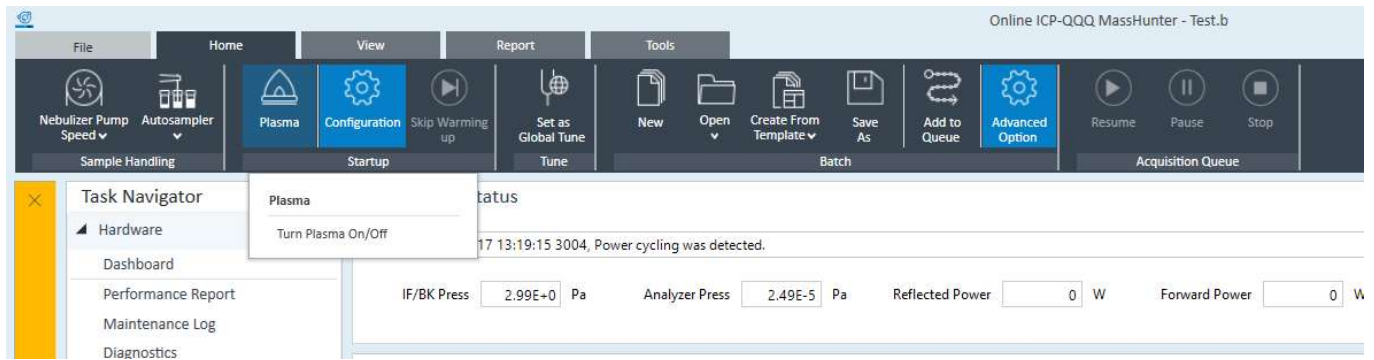


When creating a new batch from an existing batch, there are four options which carry over settings from the previous batch. For most purposes, checking the first two boxes is sufficient, which will keep the instrument settings (importantly the gas modes). Selecting "Sample List" allows the carry over of the same sample sequence if an experiment is being repeated, and QC configuration keeps the same quality control settings.

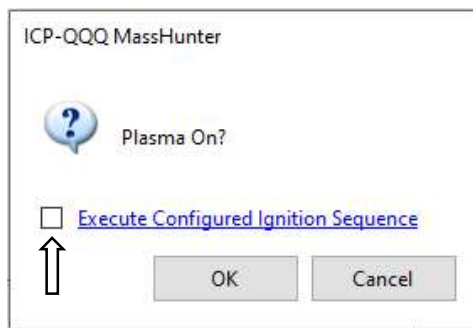


Igniting the plasma

Click on the plasma button on the top left of the screen.

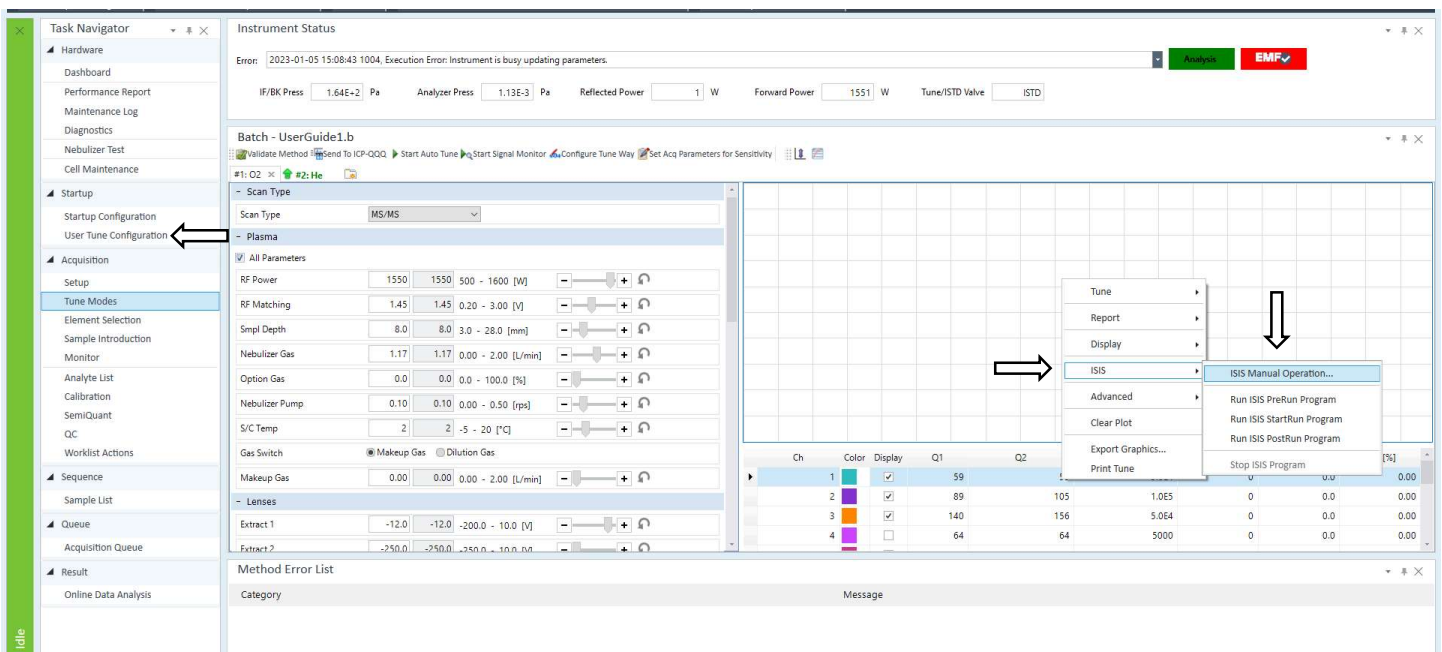


The popup window has a box that can check whether the global tune will start after the plasma has been ignited. Leave the box unchecked – the global tune will be done manually.

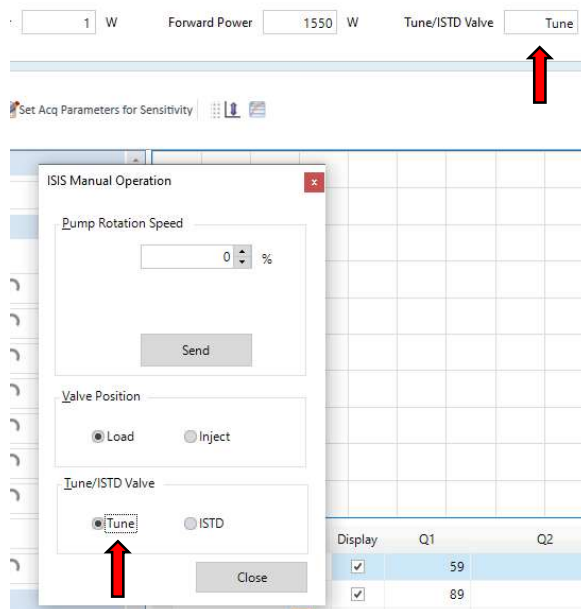


Tuning the instrument

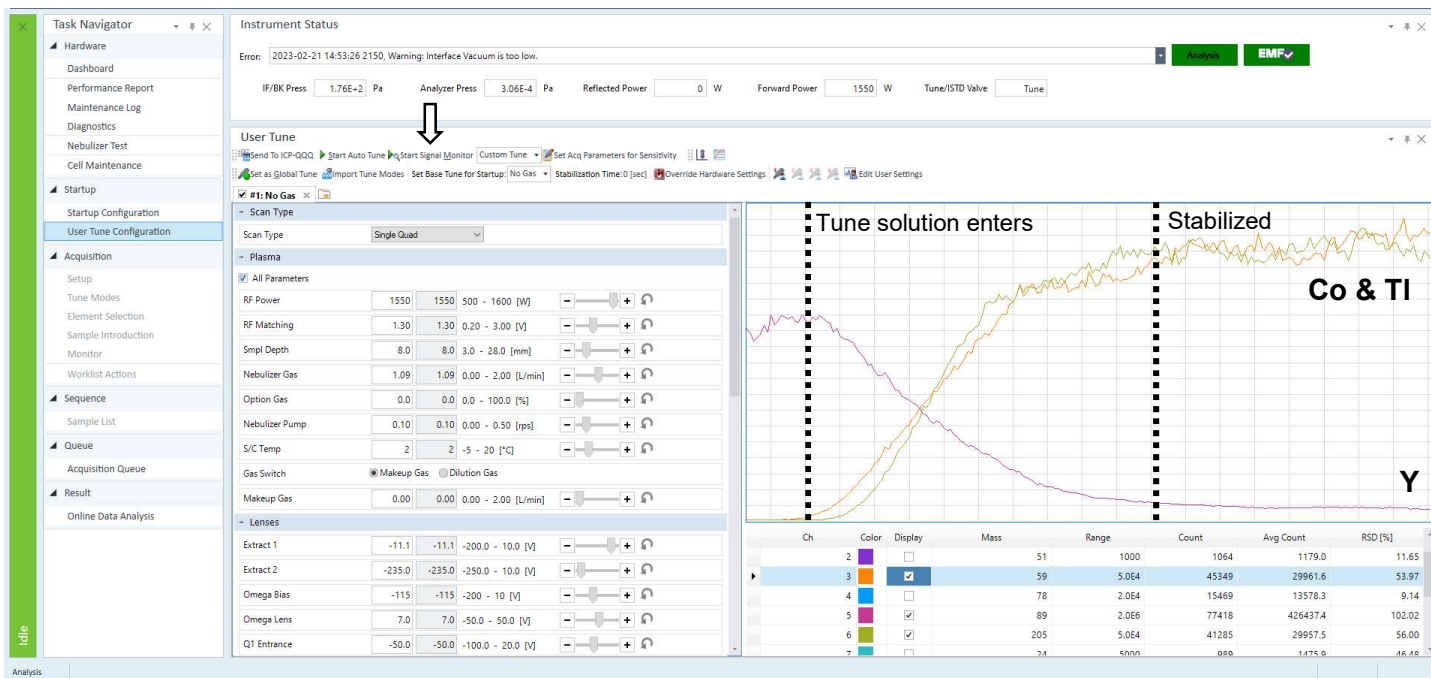
First, the valve must be set to the tune solution. Under the “Startup” section on the left side of the screen, select the “User Tune Configuration” tab, and then right click anywhere on the graph area on the right side of the screen. In the “ISIS” tab, select “ISIS Manual Operation” to bring up the valve control. *Note: Changing the valve intake can also be done in the “Tune Modes” tab of the “Acquisition” section.



In the “ISIS Manual Operation” menu, click the “Tune” button to switch the valve. The “Tune/ISTD Valve” indicator should read “Tune.” After switching, the menu can be closed.



To ensure that the tune solution is running through the plasma, select the “Start Signal Monitor” in the “User Tune Configuration” tab. The graph on the right will begin showing the counts of any selected masses. The Tune solution contains cobalt (59), cerium (140), and thallium (205). It also contains yttrium (89), but at a lower concentration than the ISTD. The signal of Co, Ce, and Tl should increase and stabilize, while Y should decrease. Once the tune solution elements have stabilized, click “Stop Signal Monitor,” which is in the same place as “Start Signal Monitor.”



Under the “Startup” section, select the “Startup Configuration” tab. For general use, check the Torch Axis, EM, Lens Tune, Resolution/Axis, and Performance Report boxes. Select the “Add Startup Tasks to Queue” to initiate the tune. This process should take 15-30 minutes, depending on if the instrument was just turned on. **Note that after this tune, the instrument will automatically switch the valve back to “ISTD.”**

Instrument Status

Error: 2022-12-17 13:19:15 3004, Power cycling was detected.

IF/BK Press Pa Analyzer Press Pa Reflected Power W Forward Power W

Startup

☒ Add Startup Tasks to Queue ☐ Select Custom Settings ☐ Configure Ignition Sequence

Hardware

Torch Axis ☒ Add to Queue

EM ☒

Plasma Correction ☐

Lens Tune ☒

Resolution/Axis ☒

Performance Report ☒

Full Spectrum ☐

P/A Factor ☐

Standard Setting

Tune/ISTD Valve Monitored Masses:

After the tune completes, the software generates a performance report as well as indicate if the tune didn't pass quality controls. The “Performance Report” tab is found under the “Hardware” section on the left.

Instrument Status

Error: 2023-01-05 15:08:43 1004, Execution Error: Instrument is busy updating parameters.

IF/BK Press Pa Analyzer Press Pa Reflected Power W Forward Power W

Mainframe -> Performance Report

☒ Run Performance Report ☒ Select Data Displayed in History View

Created Date	Performance Check	Run at End	Sensitivity			Background			Tune Parameters		
			Channel 1 Count	Channel 2 Count	Channel 3 Count	Channel 1 Count	Channel 2 Count	Channel 3 Count	Extract 1	Extract 2	Omega Bias
2023-01-05 13:26:57	Passed	No	22478.57	90630.72	54241.65	0.05	0.00	0.00	-12.0 V	-245.0 V	-100 V
2023-01-05 12:33:38	Failed	No	27029.01	114417.30	55571.47	0.10	0.00	0.00	-12.0 V	-240.0 V	-100 V
2023-01-05 11:51:31	Failed	No	27490.69	92954.89	54548.96	0.00	0.00	0.05	-11.5 V	-235.0 V	-100 V
2023-01-05 11:11:20	Passed	No	22337.49	88568.35	53377.51	0.00	0.00	0.00	-11.8 V	-245.0 V	-105 V
2022-12-21 15:52:24	Passed	No	67538.97	269522.14	176683.09	0.00	0.00	0.05	-11.8 V	-235.0 V	-100 V
2022-12-20 12:02:20	Passed	No	6207.67	32900.36	15410.22	0.00	0.00	0.00	-11.8 V	-250.0 V	-100 V
2022-12-19 10:57:00	Passed	No	18947.46	97672.28	55067.18	0.05	0.00	0.00	-12.0 V	-245.0 V	-100 V
2022-12-17 14:27:08	Passed	No	23366.24	101180.73	56289.66	0.00	0.00	0.00	-12.0 V	-245.0 V	-100 V
2022-12-15 10:23:37	Passed	No	21300.30	107551.48	59935.62	0.00	0.00	0.00	-12.0 V	-250.0 V	-100 V

Double click to view an individual performance report or select the appropriate data from the dropdown menu. If the oxides are >1.5% or doubly charged species are >2.5%, this is the likely cause of the failure. RSD% should also be >5% for all three masses.

Mainframe -> Performance Report

☒ Run Performance Report ☒ Select Data Displayed in History View

Performance **HW Settings / Tune Parameters** **Meter** **Other**

Sensitivity

Mass	7	89	205
Range	5.0E4	2.0E5	1.0E5
Count	27029.01	114417.30	55571.47
RSD%	2.3	2.7	2.6
Integration Time	0.100 sec		
Sampling Period	0.311 sec		
Oxide	156/140	2.20%	
Doubly Charged	70/140	2.94%	

Background

Mass	7	89	205
Count	0.10	0.00	0.00

Data acquisition

Acquisition configuration

Under the “Acquisition” section, select the “Setup” tab. For general use, check the “FullQuant Analysis” near the bottom. “SemiQuant Analysis” can also be checked for the instrument to give semiquantitative measurements on additional elements not selected in “Element Selection” tab. Also make sure “P/A Factor Adjustment” is checked.

Batch - UserGuide1.b

Validate Method

Acq Mode: Spectrum

Spectrum Mode Option: Q2 Peak Pattern: 1 Point

Replicates: 3

Sweeps/Replicate: 50

Acq Option:

☒ Auto/Semi Auto Tune before Batch

☒ Generate Tune Report

☒ P/A Factor Adjustment

Advanced Configuration

Set Batch Comment

Data Analysis Method	Sample Template	Batch Template
FullQuant Analysis		
Intelligent Sequence		
SemiQuant Analysis		
Isotope Ratio Analysis		
Isotope Dilution Analysis		

Analysis Mode: Spectrum

Bkg Subtraction if Exists: Count Subtraction except for ISTD

Interference Correction: Acq. Defined

Selecting tune modes

Select “Tune Modes” under the “Acquisition” section. If the batch was created from an existing batch, the parameters for each gas should have been duplicated. To remove a gas mode, click the “x” next to the gas mode name. If the “Nebulizer Gas” setting was changed in “User Tune Configuration,” ensure that value matches for each gas in “Tune Modes.” **To tune the gas modes, first switch the valve back to “Tune” (see the Tuning the Instrument section of the guide).** After the tune solution is stabilized, select “Start Auto Tune.”

Batch - UserGuide1.b

Validate Method

Send To ICP-QQQ

Start Auto Tune

Start Signal Monitor

Configure Tune Way

Set Acq Parameters for

#1: O2 x #2: He

Start Lens Auto Tune

Scan Type: MS/MS

Plasma:

All Parameters

RF Power: 1550 1550 500 - 1600 [W]

RF Matching: 1.45 1.45 0.20 - 3.00 [V]

Smpl Depth: 8.0 8.0 3.0 - 28.0 [mm]

Nebulizer Gas: 1.17 1.17 0.00 - 2.00 [L/min]

Option Gas: 0.0 0.0 0.0 - 100.0 [%]

Nebulizer Pump: 0.10 0.10 0.00 - 0.50 [rps]

S/C Temp: 2 2 -5 - 20 [°C]

Gas Switch: ☒ Makeup Gas ☐ Dilution Gas

Makeup Gas: 0.00 0.00 0.00 - 2.00 [L/min]

Lenses:

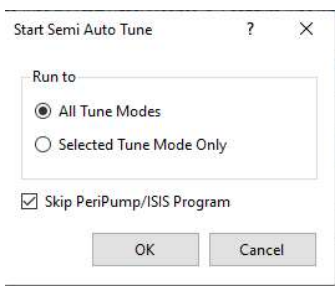
Extract 1: -12.0 -12.0 -200.0 - 10.0 [V]

Extract 2: -250.0 -250.0 -250.0 - 10.0 [V]

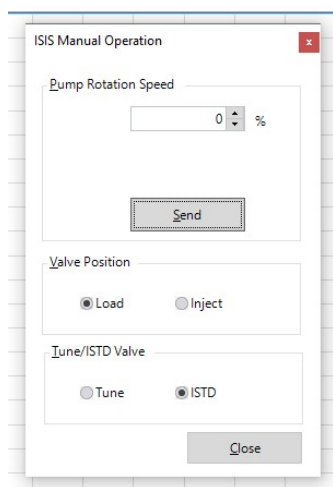
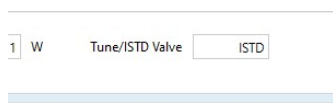
Method Error List

Category

After clicking “Start Auto Tune,” a box will pop up which allows for tuning either an individual gas mode or all gas modes. A performance report will also be generated at the end of the gas tunes. Note: O₂ oxides and doubly charged readouts will be significantly higher than other modes. The He mode will generally have lower count rates in the tune.

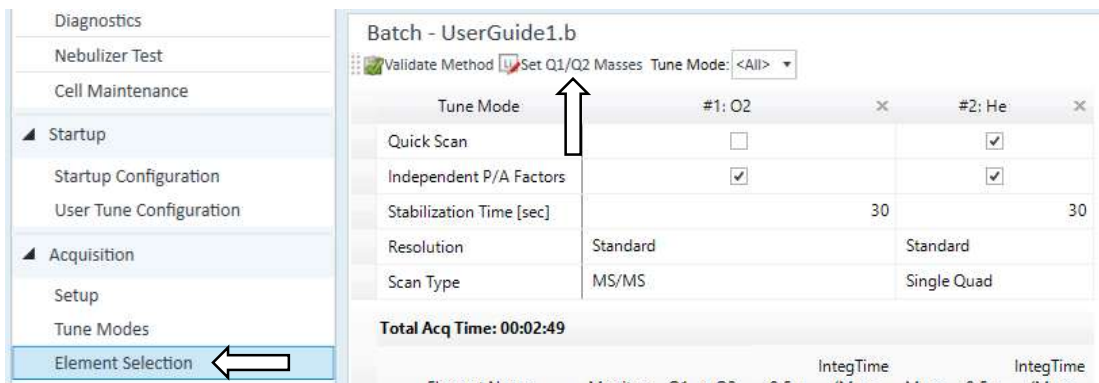


After the gas tune has completed, the instrument will NOT switch the valve back to ISTD (unlike the global tune). Opening “ISIS Manual Operation” will always automatically switch the valve back to ISTD. Be sure to switch back to ISTD before running a batch (and be sure the ISTD is stabilized).



Selecting elements

Select the “Element Selection” tab under the “Acquisition” section. To select which elements to analyze, click “Set Q1/Q2 Masses.”



A popup will show the periodic table, with elements set to be analyzed highlighted in orange. To select an element for analysis, click that element on the periodic table, then at the bottom, select individual masses to monitor. Be sure to also select elements for the internal standard, which include Li⁶, Sc, Y, In, Bi, and Tb. It is recommended to choose internal standard elements which are close in mass to elements of interest. To switch between gas modes, click the dropdown menu at the top.

Select Elements on Periodic Table

Precursor Ion Scan... Product Ion Scan... Set Mass Shift...

Tune Mode: He

Selected Elements: 21

Mass	Element
23	Na
24	Mg
45	Sc
55	Mn
56	Fe
57	Fe
58	Ni
59	Co
60	Ni
63	Cu
64	Zn
65	Cu
66	Zn
75	As
77	Se

Go to Mass Scale Element Information...

Correction Equations: 0

OK Cancel

Certain gas modes cause the target mass to shift to a different mass. When selecting elements in these gas modes (for general use, this will be O₂), check the appropriate shift on the right. Also be sure to select shifts for internal standard elements as well. Once all desired elements have been selected in every gas mode, click "OK" in the bottom right corner.

Select Elements on Periodic Table

Precursor Ion Scan... Product Ion Scan... Set Mass Shift...

Tune Mode: O2

Mass Pairs: 6

Element	Q1	Q2	Mass Shift
P	31	47	16
S	32	48	16
Sc	45	61	16
As	75	91	16
Se	78	94	16
Y	89	105	16

Set Mass Shift

☐ +0 (On-Mass)

Predefined Shift

☐ +15 (NH)

☒ +16 (NH2 or O)

☐ +17 (NH3)

☐ +18 (NH4)

NH3 Cluster

Dimer Trimer Tetramer Pentamer

☐ +32 (NH(NH3))

☐ +33 (NH2(NH3))

☐ +34 ((NH3)2)

Custom Shift

Update

Custom shift Masses and/or Mass ranges separated by commas. For example, type 1,18, 28-30. After entering the custom shift, click Update button or hit enter to apply them to the Mass Pairs table on the left.

OK Cancel

After confirming all desired elements, set the integration times for each element for each gas mode in the table in “Element Selection.” For general use, an integration time of 0.6s (3s in oxygen mode) is sufficient for many elements. Note that an element with a mass shift is treated as a separate entry. Check monitor to show realtime measurements of the element. “+0.5 u” should not be checked under most circumstances.

Batch - UserGuide1.b

Validate Method Set Q1/Q2 Masses Tune Mode: <All>

Tune Mode	#1: O2	#2: He
Quick Scan	☐	☒
Independent P/A Factors	☒	☒
Stabilization Time [sec]	30	30
Resolution	Standard	Standard
Scan Type	MS/MS	Single Quad

Total Acq Time: 00:02:49

Element Name	Monitor	Q1 -> Q2	+0.5 u	IntegTime /Mass [sec]	Mass	+0.5 u	IntegTime /Mass [sec]
Na	☐	N/A	☐	N/A	23	☐	0.6000
Mg	☐	N/A	☐	N/A	24	☐	0.6000
P	☒	31 -> 47	☐	3.0000	N/A	☒	
S	☒	32 -> 48	☐	3.0000	N/A	☒	
K	☐	N/A	☐	N/A	39	☐	0.6000
Ca	☐	N/A	☐	N/A	44	☐	0.6000
Sc	☐	N/A	☐	N/A	45	☐	0.6000
Sc	☒	45 -> 61	☐	0.9999	N/A	☒	
V	☐	N/A	☐	N/A	51	☐	0.6000
Cr	☐	N/A	☐	N/A	52	☐	0.6000

After setting the elements in “Element Selection,” go to the “Analyte List” tab under the “Acquisition” section. If the batch was created from an existed batch and no elements were added or subtracted, then no changes are needed. However, if a change was made, it is easiest to select the entire table (click the upper left box) and remove all the entries by clicking the “X” at the top. After clearing the table, click the “Load List from Acquisition Method” button at the top to repopulate the list.

Diagnostics

Nebulizer Test

Cell Maintenance

Startup

Startup Configuration

User Tune Configuration

Acquisition

Setup

Tune Modes

Element Selection

Sample Introduction

Monitor

Analyte List

Calibration

SemiQuant

QC

Worklist Actions

Sequence

Sample List

Queue

Acquisition Queue

Batch - UserGuide1.b

Validate Method DA Method Task: X

Load List From Acquisition Method

Tune Mode	Scan Type	Q1 -> Q2	Mass	P/A	Mass	Analyte/ISTD
1: O2	MS/MS	31 -> 47	31	P	47	Analyte
2: O2	MS/MS	32 -> 48	32	S	48	Analyte
3: O2	MS/MS	45 -> 61	45	Sc	61	Analyte
4: O2	MS/MS	89 -> 105	89	Y	105	Analyte
2: He	Single Quad			Na	23	Analyte
2: He	Single Quad			Mg	24	Analyte
2: He	Single Quad			Sc	45	Analyte
2: He	Single Quad			Mn	55	Analyte
2: He	Single Quad			Fe	56	Analyte
2: He	Single Quad			Fe	57	Analyte
2: He	Single Quad			Ni	58	Analyte
2: He	Single Quad			Co	59	Analyte
2: He	Single Quad			Ni	60	Analyte
2: He	Single Quad			Cu	63	Analyte
2: He	Single Quad			Zn	64	Analyte
2: He	Single Quad			Cu	65	Analyte
2: He	Single Quad			Zn	66	Analyte
2: He	Single Quad			As	75	Analyte

After loading the list from acquisition method, designate internal standard elements by changing the entry in a dropdown under the Analyte/ISTD column. Note that internal standards need to be set for each gas mode (i.e. setting Sc to an internal standard in O₂ does not set it in He).

Batch - UserGuide1.b

Validate Method

DA Method Task:

Analyte							
	Tune Mode	Scan Type	Transition	Q1	Name	Q2	Analyte/ISTD
1	1: O2	MS/MS	31 -> 47	31	P	47	Analyte
2	1: O2	MS/MS	32 -> 48	32	S	48	Analyte
3	1: O2	MS/MS	45 -> 61	45	Sc	61	Analyte
4	1: O2	MS/MS	89 -> 105	89	Y	105	Analyte
5	2: He	Single Quad			Na	23	ISTD
6	2: He	Single Quad			Mg	24	Analyte
7	2: He	Single Quad			Sc	45	Analyte
8	2: He	Single Quad			Mn	55	Analyte
9	2: He	Single Quad			Fe	56	Analyte
10	2: He	Single Quad			Fe	57	Analyte
11	2: He	Single Quad			Ni	58	Analyte
12	2: He	Single Quad			Co	59	Analyte
13	2: He	Single Quad			Ni	60	Analyte
14	2: He	Single Quad			Cu	63	Analyte
15	2: He	Single Quad			Zn	64	Analyte
16	2: He	Single Quad			Cu	65	Analyte
17	2: He	Single Quad			Zn	66	Analyte
18	2: He	Single Quad			As	75	Analyte

Setting calibration standards

Select the “Calibration” tab under the “Acquisition” section. For general use, set the “Curve Fit” to Linear and “Origin” to Blank Offset in the respective dropdown menus. For “ISTD,” generally set the internal standard element closest in mass to the analyte element. “Units” should be set to the relevant concentration range of the samples, which will likely be ppb.

Diagnostics

Nebulizer Test

Cell Maintenance

Startup

Startup Configuration

User Tune Configuration

Acquisition

Setup

Tune Modes

Element Selection

Sample Introduction

Monitor

Analyte List

Calibration

SemiQuant

QC

Batch - UserGuide1.b

Validate Method

DA Method Task:

Basic Calibration Parameters

Calibration Title

Calibration Method

Edit ISTD Conc

Weighting

Virtual ISTD Correction

External Calibration

	Tune Mode	Transition	Q1	Name	Q2	Curve Fit	Origin	ISTD	Min Conc.	Units	Outlier	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7
1	1: O2	31 -> 47	31	P	47	Linear	Blank offset	45 -> 61	0		☒	0	0.1	1	10	100	1000	
2	1: O2	32 -> 48	32	S	48	Linear	Blank offset	<None>	0		☒							
3	2: He			Na	23	Linear	Blank offset	45 -> 61	0		☒							
4	2: He			Mg	24	Linear	Blank offset	89 -> 105	0		☒							
5	2: He			Mn	55	Linear	Blank offset	45	0		☒							
6	2: He			Fe	56	Linear	Blank offset	45	0		☒							
7	2: He			Fe	57	Linear	Blank offset	45	0		☒							
8	2: He			Ni	58	Linear	Blank offset	45	0		☒							

For each element, the concentration in each standard (Level) needs to be set. Level 1 should be 0 for all elements. If each element is the same concentration within a standard, click and drag to highlight all the levels for all the elements, then right click and select fill down. If the concentrations vary between elements within a standard, they need to be set manually. For general use, the QC and spike columns do not need to be filled in.

Instrument Status

Error: 2023-01-05 15:08:43 1004, Execution Error: Instrument is busy updating parameters.

IF/BK Press

1.65E+2

Pa

Analyzer Press

1.13E-3

Pa

Reflected Power

0

W

Forward Power

1551

W

Tune/ISTD Valve

Batch - UserGuide1.b

Validate Method

DA Method Task:

Basic Calibration Parameters

Calibration Title

Calibration Method

Edit ISTD Conc

Weighting

Virtual ISTD Correction

External Calibration

Analyte											Level				
Tune Mode	Transition	Q1	Name	Q2	Curve Fit	Origin	ISTD	Min Conc.	Units	Outlier	Level 1	Level 2	Level 3	Level 4	Level 5
13	2: He		Cu	65	Linear	Blank offset		45	0		0	0.1	1	10	100
14	2: He		Zn	66	Linear	Blank offset		45	0		0	0.1	1	10	100
15	2: He		As	75	Linear	Blank offset		89	0		0	0.1	1	10	100
16	2: He		Se	77	Linear	Blank offset		89	0		0	0.1	1	10	100
17	2: He		Se	78	Linear	Blank offset		89	0		0	0.1	1	10	100
18	2: He		Cd	112	Linear	Blank offset		115	0		0	0.1	1	10	100
19	2: He		Cd	114	Linear	Blank offset		115	0		0	0.1	1	10	100

Export Table...

Export Selected Area...

Copy

Paste

Add Levels

Delete Level

Delete

Reset Sort

Lock Analyte Columns

Fill Down

Fill Across

Multiply Conc...

Outlier Settings

Page Setup...

Print...

Print Preview...

Blank

Spike Amount

5

BlkVrfy

Spike1

Spike2

Spike3

Setting up a sequence

Click the “Sample List” tab in the “Sequence” section. To being creating a sample entry, click the dropdown under “Sample Type.” There are many options, but for general use CalBlk, CalStd, and Sample will be the main identifiers.

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Startup Configuration

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Sample Introduction

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Analyte List

Calibration

SemiQuant

QC

Worklist Actions

Sequence

Sample List

Queue

Acquisition Queue

Result

Instrument Status

Error: 2023-01-05 15:08:43 1004, Execution Error: Instrument is busy updating parameters.

IF/BK Press

1.65E+2

Pa

Analyzer Press

1.13E-3

Pa

Reflected Power

0

W

Forward Power

Batch - UserGuide1.b

Validate Method

Use Task List

Import Sample List

Estimated Time for Batch Acquisition: 00:00:00

Skip

Sample Type

Sample Name

Vial#

Level

File Name

Comment

1

2

Bkgnd

3

BlkVrfy

4

CalBlk

5

CalStd

6

CICSpk

7

DilStd

8

DriftChk

9

FQBlk

10

IsoStd

11

QC1

12

QC2

13

QC3

14

QC4

15

QC5

16

Sample

17

Spike

18

Spike Ref

19

SQBlk

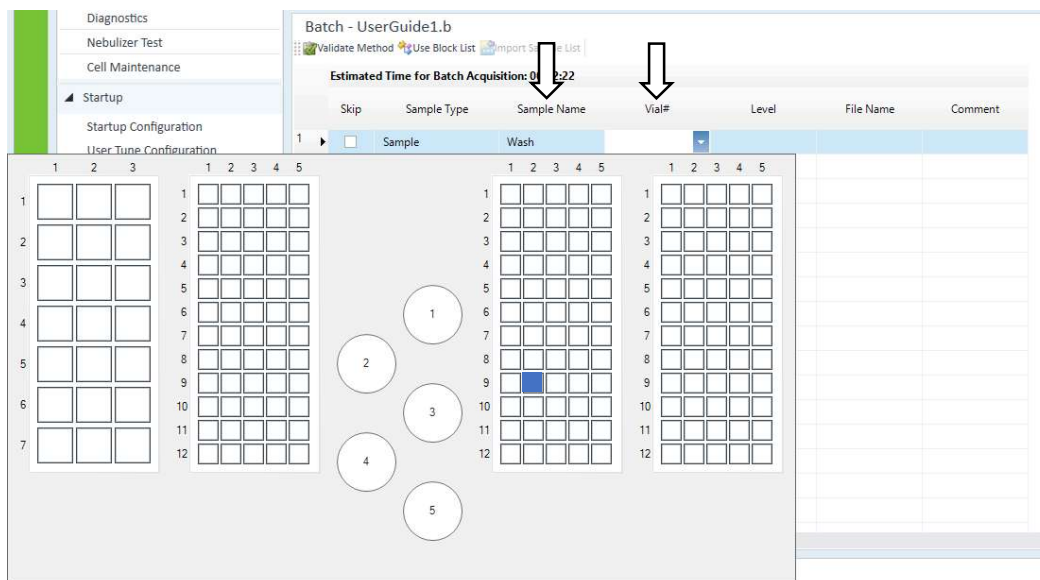
20

SQISTD

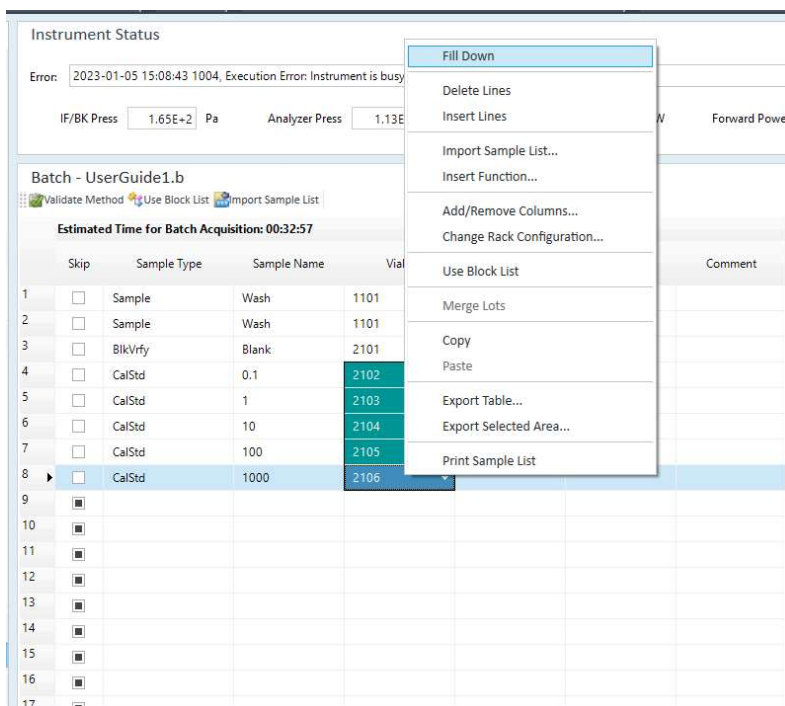
21

SQStd

After selecting the sample type, select the position of the sample in the autosampler. Clicking the dropdown will bring up a diagram of the autosampler. Clicking the appropriate position will autofill the location in the autosampler. The syntax for the racks (minus the middle column) is 4 digits, with the first digit being the rack number, the second digit being the column number, and the last two digits being the vial number in that column (e.g. 3209 would be rack 3, column 2, vial 9, highlighted in blue in the picture below). The sample can also be labeled under the “Sample Name” column.



The Fill Down function can be used in several ways in the sample list. Highlighting a set of samples under “Vial” and using Fill Down (right click on the highlighted section and select Fill Down) will set the positions in the autosampler sequentially, starting from the topmost vial in the selection. If sample names have an integer value, Fill Down will iteratively add 1 to each subsequent sample (e.g. Sample1, Sample2...). For “Sample Type,” Fill Down instead copies the type in the top entry in the highlight and applies that same type to all the samples in the highlight.



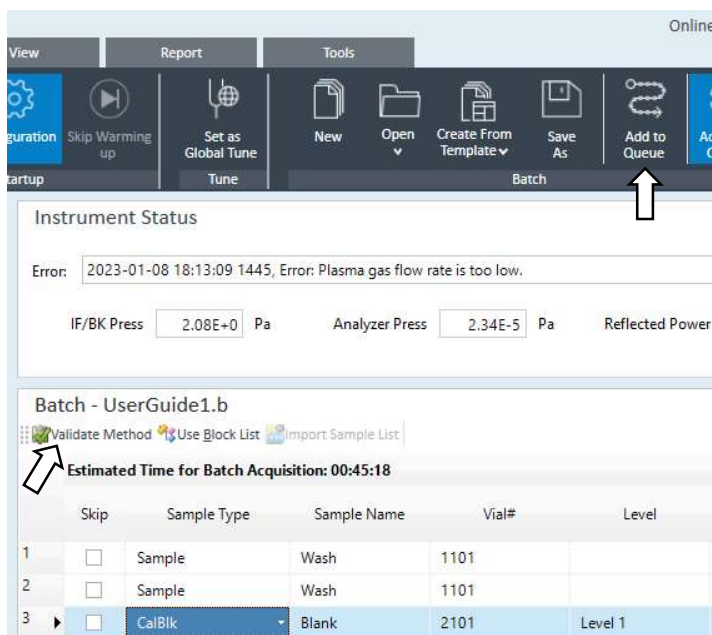
General sequence design

Besides running samples, there are two important measurements for each batch: the blank and the calibration curve. The blank sets the baseline for the elemental content of a matrix. Generally, a few washes should precede the blank to ensure there are no residual material. Both your washes and blank should be the same matrix as your samples. The blank is the CalBlk sample type and should be set to the same level with 0 for concentration (generally Level 1). The blank can generally be followed by the calibration standards to instruct the instrument that a specific detector response correlates to a specific concentration of an element. The standards should ideally have a range in concentrations that all samples fall between. The CalStd sample type is used for standards, and make sure to also put the appropriate level for each standard. Finally, it is good practice to have washes between every few samples to prevent accumulation of material that would elevate the background. Depending upon how concentrated the samples are, this can range from every 20 samples to after every sample, and perhaps even multiple washes between each.

	Skip	Sample Type	Sample Name	Vial#	Level
1	☐	Sample	Wash	1101	
2	☐	Sample	Wash	1101	
3	☐	CalBlk	Blank	2101	Level 1
4	☐	CalStd	0.1	2102	Level 2
5	☐	CalStd	1	2103	Level 3
6	☐	CalStd	10	2104	Level 4
7	☐	CalStd	100	2105	Level 5
8	☐	CalStd	1000	2106	Level 6
9	☐	Sample	Wash	1101	
10	☐	Sample	Check100	2105	
11	☐	Sample	Wash	1101	

Running the batch

Once the sequence is ready to be run, it is advisable to click “Validate Method,” which will detect if there are any inconsistencies in the settings (i.e. Element Selection not matching Analyte List). Saving the batch will also automatically do a validation check. To start running the instrument, click “Add to Queue” at the top.



Online

View | Report | Tools

Duration | Skip Warming up | Set as Global Tune | New | Open | Create From Template | Save As | Add to Queue | Add to Queue

Startup | Tune | Batch

Instrument Status

Error: 2023-01-08 18:13:09 1445, Error: Plasma gas flow rate is too low.

IF/BK Press 2.08E+0 Pa Analyzer Press 2.34E-5 Pa Reflected Power

Batch - UserGuide1.b

Validate Method | Use Block List | Import Sample List

Estimated Time for Batch Acquisition: 00:45:18

	Skip	Sample Type	Sample Name	Vial#	Level
1	☐	Sample	Wash	1101	
2	☐	Sample	Wash	1101	
3	☐	CalBlk	Blank	2101	Level 1

Data analysis

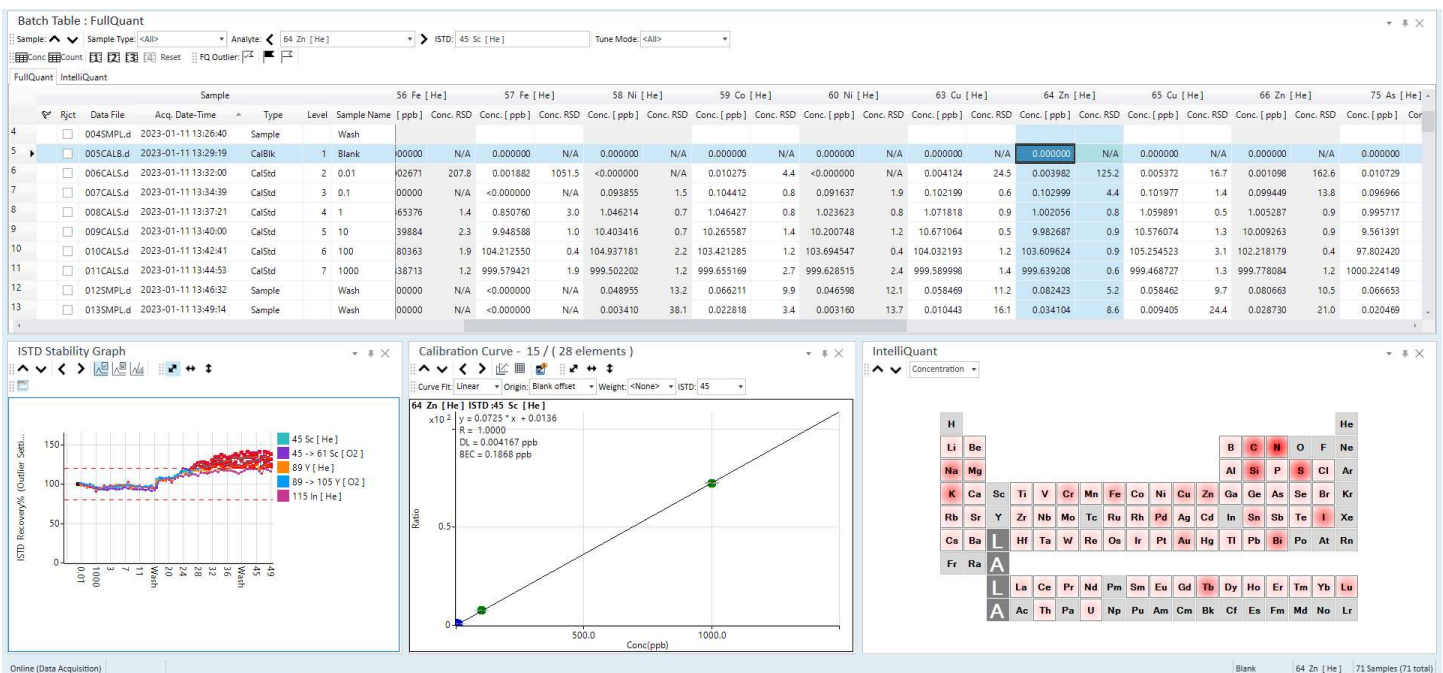
Real-time data analysis

Once the sequence is initiated, a separate program will open that will report data as it is being collected by the instrument, as well as allow for some real-time processing of the data. The data can be reported in either raw counts, or once the standard curve is generated, concentration, which can be changed by clicking “Conc” or “Count” in the upper left of the graph. Other features to note – the ISTD Stability Graph, Calibration Curve, and if SemiQuant was selected during setup, the IntelliQuant.

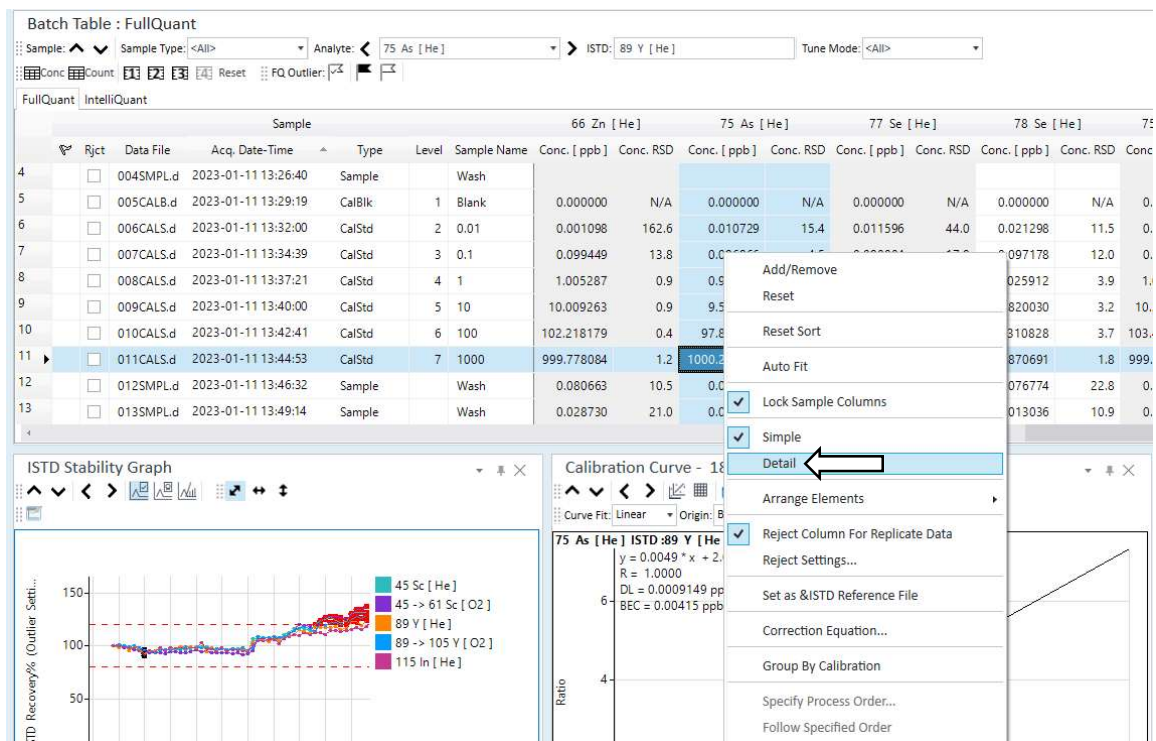
The ISTD Stability Graph shows how much the internal standard signal is changing compared to the blank. If the internal standards begin to fluctuate significantly (as shown below), this may suggest there are matrix effects impacting the signal.

The Calibration Curve will indicate how linear the detector responds to changes in concentration. Ideally, the R^2 value should be as close to 1 as possible. Furthermore, the BEC value is a rough indicator of the detection limit of the selected element for a particular run.

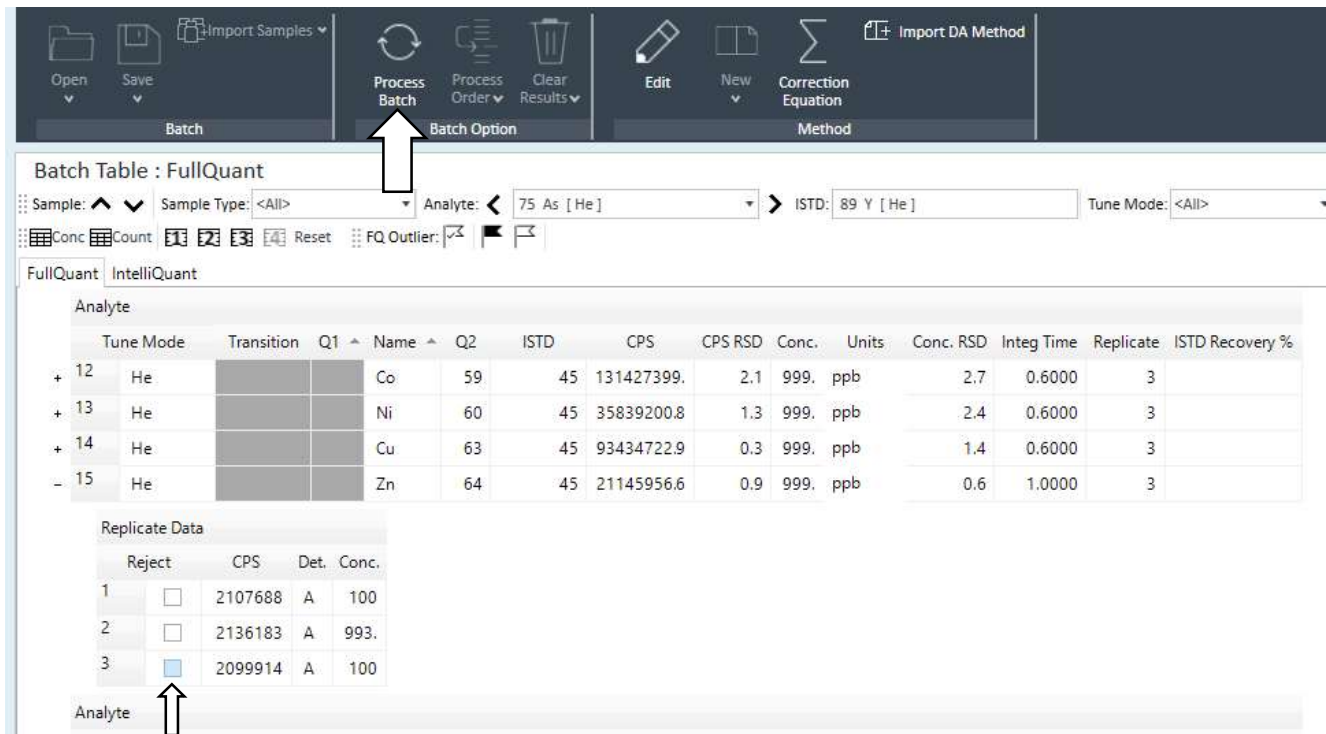
The IntelliQuant section will show a heatmap of detected elements if SemiQuant was selected during setup.



Individual reads can be investigated in more detail by right clicking on the data point and selecting “Detail.”



Clicking the “+” next to the appropriate element in the data point. If there is a suspected error in a read, it can be processed out by checking the box under “Reject,” then clicking on “Process Batch” at the top. Note that processing data **should only be done if there is sufficient and compelling evidence** that the data point has a confounding error (i.e. running out of sample).



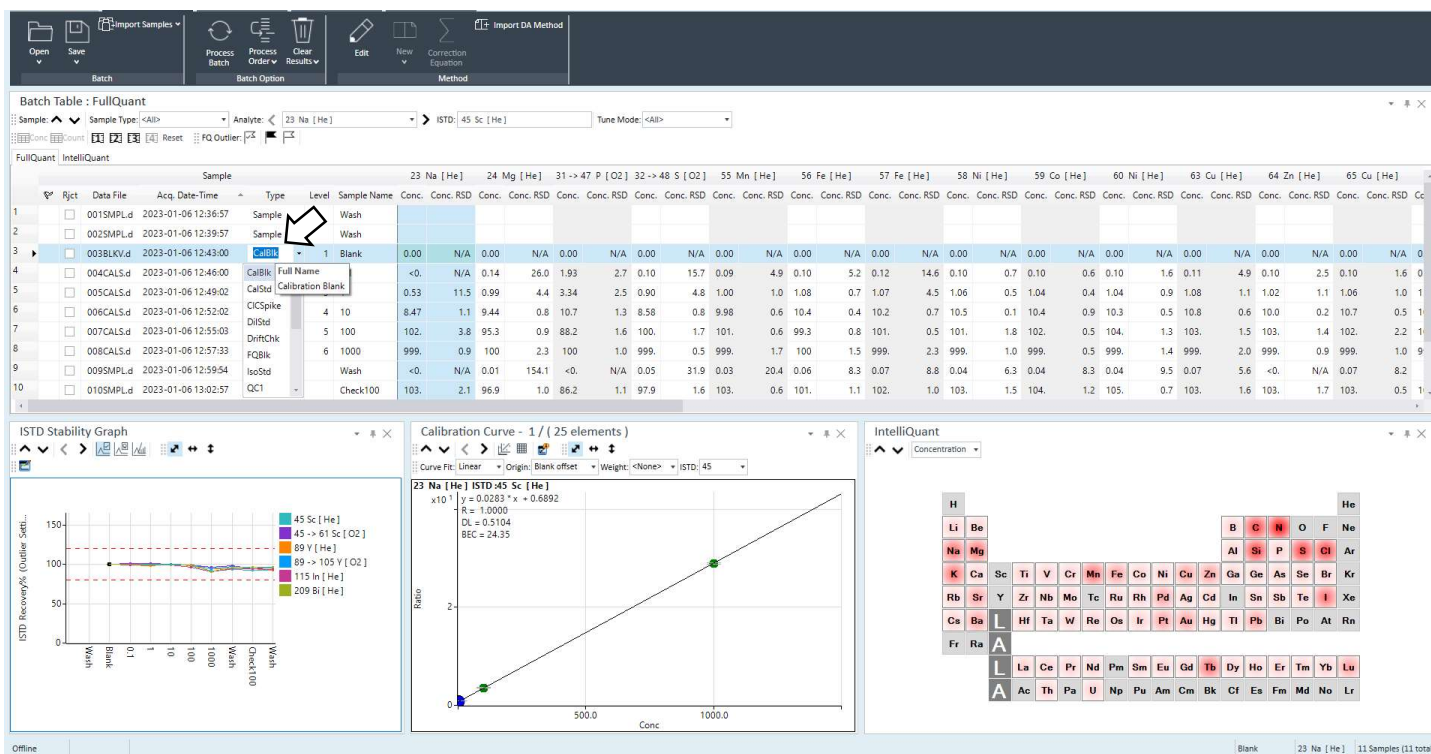
Offline data analysis

Data can also be analyzed without turning on the instrument. Click “Offline Data Analysis” on the desktop.



The offline data analysis software functions similarly to the online software. A few additional pointers for processing data:

Sample types can be changed by clicking the dropdown under “Type” in the event a sample type was inappropriately assigned during the batch.



The element for internal standard correction can also be changed by clicking the ISTD dropdown in the “Calibration Curve” section. This might be necessary if the internal standard is not stable throughout the experiment or if the sample contains that internal standard.

